

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

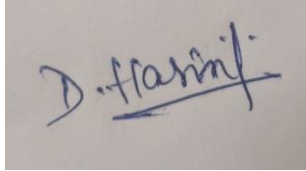
1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
5. Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 1. Find the predicted output.
 2. Calculate the error.
 3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I D.Hasini-192425147 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in blue ink, appearing to read 'D. Hasini', is written on a light-colored rectangular background.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. Maximum Likelihood Estimation (MLE) for Coin Toss

Problem Understanding

A coin is tossed **10 times**.

Outcomes:

H H T H H H T H H T

Number of Heads = 7

Number of Tails = 3

We need to estimate the probability of getting Heads (**p**) using MLE.

Method Selection

For a Bernoulli/Binomial experiment,

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Trials}}$$

where

- Success = Head
- Total Trials = 10

Solution Process

$$\hat{p} = \frac{7}{10} = 0.7$$

Accuracy of Calculation

Estimated probability of Head:

$$\boxed{\hat{p} = 0.7}$$

Interpretation of Results

The Maximum Likelihood Estimate shows that the probability of getting a Head is **0.7 (70%)**, indicating that the coin produced Heads more frequently in this experiment.

2. MLE for Students Passing an Exam

Problem Understanding

Total students = 50

Students passed = 42

Estimate the probability of passing.

Method Selection

MLE Formula:

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Observations}}$$

Solution Process

$$\begin{aligned}\hat{p} &= \frac{42}{50} \\ \hat{p} &= 0.84\end{aligned}$$

Accuracy of Calculation

Estimated probability:

$$\boxed{\hat{p} = 0.84}$$

Interpretation of Results

The estimated probability that a student passes the exam is **84%**, indicating a high success rate among the students.

3. MLE for Defective Light Bulbs

Problem Understanding

Sample size = **100 bulbs**

Defective bulbs = **8**

Estimate the probability of a bulb being defective.

Method Selection

MLE Formula:

$$\hat{p} = \frac{\text{Number of Defective Bulbs}}{\text{Total Bulbs}}$$

Solution Process

$$\begin{aligned}\hat{p} &= \frac{8}{100} \\ \hat{p} &= 0.08\end{aligned}$$

Accuracy of Calculation

Estimated probability:

$$\boxed{\hat{p} = 0.08}$$

Interpretation of Results

The probability that a bulb is defective is estimated to be **0.08 (8%)**, which means approximately **8 out of every 100 bulbs** are expected to be defective.

4. MLE for Rolling a 6 on a Biased Die

Problem Understanding

Total rolls = **60**

Number of sixes = **18**

Estimate the probability of rolling a 6.

Method Selection

MLE Formula:

$$\hat{p} = \frac{\text{Number of Sixes}}{\text{Total Rolls}}$$

Solution Process

$$\begin{aligned}\hat{p} &= \frac{18}{60} \\ \hat{p} &= 0.30\end{aligned}$$

Accuracy of Calculation

Estimated probability:

$$\boxed{\hat{p} = 0.30}$$

Interpretation of Results

The estimated probability of rolling a **6** is **0.30 (30%)**, indicating that the die is biased since this probability is greater than the fair-die probability of $1/6 \approx 0.167$.

5. Prediction, Error and Weight Update

Problem Understanding

Given:

Input

$$x = 4$$

Weight

$$w = 2$$

Actual Output

$$y_{actual} = 10$$

Prediction Formula

$$y = w \times x$$

Gradient

$$= 4$$

Learning Rate

$$\alpha = 0.01$$

Find:

1. Predicted Output
2. Error
3. Updated Weight

Method Selection

Prediction:

$$y = w \times x$$

Error:

$$\text{Error} = y_{actual} - y_{predicted}$$

Weight Update:

$$w_{new} = w - \alpha \times \text{Gradient}$$

Solution Process

Step 1: Predicted Output

$$y = 2 \times 4 = 8$$

Predicted Output:

$$\boxed{8}$$

Step 2: Error

$$\text{Error} = 10 - 8 = 2$$

$$\boxed{\text{Error} = 2}$$

Step 3: Updated Weight

$$\begin{aligned} w_{new} &= 2 - (0.01 \times 4) \\ &= 2 - 0.04 \\ &= 1.96 \end{aligned}$$

Updated Weight:

$$\boxed{1.96}$$

Accuracy of Calculation

- Predicted Output = **8**
- Error = **2**
- Updated Weight = **1.96**
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Interpretation of Results

The model predicts an output of **8**, while the actual output is **10**, resulting in an error of **2**. After

applying gradient descent with a learning rate of **0.01**, the weight decreases from **2** to **1.96**, helping the model learn and improve its predictions in future iterations.